

En Masse Pharmaceutical Product Bans

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Do political statements without a scientific basis on medicines cause global pharmaceutical demand spillovers during pandemics? We examine this question using novel sales data from 2018-2020 in the Indian domestic anti-malarial drug markets. Our context is the President of the United States (POTUS) Donald Trump's exogenous endorsement of the anti-malarial drug hydroxychloroquine (HCQ) during the ongoing Covid-19 pandemic. We find that between March and June 2020, after the POTUS endorsement, HCQ and congener medicine chloroquine (CQ) sales in India increased by 70.15% and 28.1% respectively compared to the average of same period in 2019 and 2018. Overall pharmaceutical sales for other anti-malarial drugs (our control group) experienced a decline (probably due to potential crowding-out or shortages). We next utilize a difference-in-difference setup coupled with an event-study design framework, to exploit the endorsement by the POTUS as an exogenous shock. We find supporting causal evidence that revenue experienced a growth rate of 1.1% and 0.67% for HCQ and CQ respectively, while the analogous figures for sales were 0.96% and 0.67% after controlling for molecule, region, and time specific unobserved heterogeneity. Also, heterogeneity analysis show that regions comprising higher share of urban population respond sharply vis-a-vis other regions, mainly because they possess better digital infrastructure which coupled with their high population density resulted in social-media induced network effects. Our findings, probably the first globally on the cross-border market effects of POTUS endorsement of HCQ have normative health policy implications for future global pandemics and associated misinformation around medicine and politics.

I. Introduction